

VINAY YALAMANCHI

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CAREER OBJECTIVE:

AI/ML Engineer and Computer Science Master's graduate with 3+ years of experience designing, developing, testing, and deploying scalable software applications, REST APIs, and AI-enabled solutions using Python, C#, FastAPI, SQL, PostgreSQL, and Microsoft Azure. Skilled in software development, object-oriented programming, debugging, software testing, CI/CD automation, Agile methodologies, and machine learning integration. Proven ability to build production-ready applications, optimize system performance, and collaborate effectively within cross-functional engineering teams. Strong foundation in software engineering principles, DevOps practices, cloud technologies, and problem-solving.

SUMMARY OF SKILLS

- **Programming Languages:** Python, C#, JavaScript, SQL
- **Software Engineering:** Object-Oriented Programming (OOP), REST APIs, Data Structures, Algorithms, SDLC, Software Testing, Unit Testing, Debugging, Code Reviews
- **Frameworks & Technologies:** FastAPI, React, HTML5, CSS3
- **Databases:** PostgreSQL, MySQL
- **Cloud & DevOps:** Microsoft Azure, Azure DevOps, CI/CD Pipelines, Docker, Git, GitHub
- **Machine Learning & AI:** Scikit-Learn, TensorFlow, PyTorch, Hugging Face Transformers, Generative AI, Feature Engineering, Model Evaluation, Model Deployment
- **Methodologies:** Agile Development, Scrum, Design Thinking, Hypothesis Driven Development
- **Productivity Tools:** Microsoft Excel, Word, PowerPoint

PROFESSIONAL EXPERIENCE

UL Solutions | AI Engineer Fremont, California

Mar 2026 – Present

- Develop AI-enabled applications and backend services using Python, FastAPI, SQL, and cloud technologies.
- Design and maintain APIs, automation workflows, and data-processing solutions supporting engineering operations.
- Collaborate with cross-functional teams in Agile environments to deliver scalable software solutions.
- Implement CI/CD pipelines, testing, debugging, and deployment processes to improve software reliability.
- Integrate machine learning and data-driven capabilities into enterprise applications.
- Contribute to software design, code reviews, and technical documentation following engineering best practices.

Fractal Analytics | AI Engineer Bangalore, India

Feb 2023 – Nov 2023

- Developed AI-powered applications and backend services using Python, C#, FastAPI, and REST APIs to support enterprise automation initiatives.
- Built scalable API integrations and microservices enabling real-time processing of customer and business data across multiple workflows.
- Implemented Generative AI capabilities using Hugging Face Transformers and prompt-engineering techniques to automate content-generation processes.
- Collaborated with software engineers, product managers, and analysts in Agile environments to deliver high-quality production-ready solutions.
- Configured Azure DevOps CI/CD pipelines to automate testing, deployment, and release management processes, improving deployment reliability.
- Performed debugging, code reviews, and performance optimization activities to improve application stability, maintainability, and scalability.

Fractal Analytics | Machine Learning Engineer

Sep 2021 – Jan 2023

- Designed and developed machine learning solutions using Python and Scikit-Learn, improving predictive accuracy by 12% through feature engineering and model optimization.
- Performed data preprocessing, feature engineering, and model evaluation on large datasets to improve model performance and training efficiency.
- Developed deep learning prototypes using TensorFlow and PyTorch while applying software engineering best practices for maintainable and reusable code.
- Built automated model deployment workflows using Microsoft Azure and Azure DevOps, streamlining production releases and monitoring.

- Collaborated with developers, analysts, and stakeholders in Agile teams to deliver data-driven software solutions aligned with business goals.
- Authored technical documentation, testing procedures, deployment guides, and validation reports to support ongoing maintenance and knowledge transfer.

Claro Software Solutions Pvt Ltd |Software Engineering Intern

Feb 2021 – Aug 2021

- Developed responsive web applications using HTML, CSS, and JavaScript while ensuring cross-browser compatibility and mobile responsiveness.
- Implemented user authentication, order tracking, and payment-processing functionalities to enhance application capabilities and user experience.
- Collaborated with development teams to gather requirements, implement features, and resolve technical issues in Agile environments.
- Participated in software debugging, testing, and performance optimization activities to improve application reliability and functionality.
- Assisted in sprint planning, stand-ups, peer reviews, and feature validation as part of the software development lifecycle.
- Applied Design Thinking and Hypothesis Driven Development methodologies to improve usability and overall customer satisfaction.

PROJECT’S

1. AI-Assisted Interview Simulator

- Designed and developed an AI-Assisted interview simulator that conducts role-specific mock interviews.
- Implemented speech-to-text (STT) to capture spoken answers and semantic similarity scoring to evaluate relevance.
- Integrated analytics to measure filler words, pauses, and confidence scoring, providing structured feedback for interviewees.
- Built dashboards for interviewees and admins using Python, Streamlit, Hugging Face, Torch, Transformers, and PostgreSQL for data storage.

2. Block chain for the Trustworthy Decentralized Web Hosting

- Built a decentralized web hosting platform using block chain technology.
- Ensured data integrity and transparency across all users through block chain protocols.

3. End-to-End Conversion Speed Analysis of an FPT.AI-based Text-to-Speech Application

- Conducted performance analysis of text-to-speech conversion times using the FPT.AI API.
- Optimized data flow between local and remote servers to reduce processing time.

4. Internet of Things Projects: Smart Agriculture & Home Automation

EDUCATION

➤ Master of Science in Computer Science, Rivier University, USA	-	Jan 2024 – Dec 2025
➤ Bachelor of Technology in Computer Science and Engineering, Jawaharlal Nehru Technological University, MIST	-	2018 – 2022

CERTIFICATIONS

➤ Data Analysis with R Programming	-	Google (May 2023)
➤ Python for Data Science, AI & Development	-	IBM (Jun 2023)
➤ Data Structures and Algorithms	-	Udemy (March 2025)
➤ Claude Platform 101	-	Anthropic (2026)
➤ Claude Code in Action	-	Anthropic (2026)
➤ Claude with the Anthropic API	-	Anthropic (2026)